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Compounds **1** and **2** are modified L-amino acids. They were coupled to form Compound **3**, a dipeptide, which was then deprotected to yield Compound **4** (Figure 1).

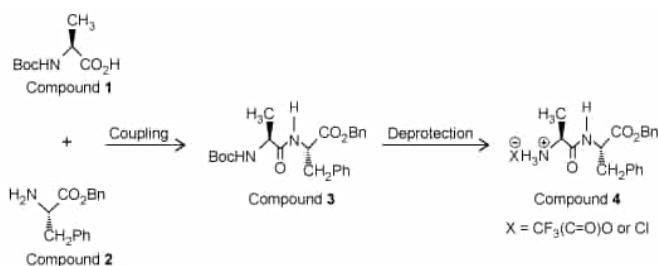


Figure 1 Synthesis of Compound **4**

To determine the effect that solvent type and temperature have on the synthesis of this peptide, the nontoxic solvents PC and EC as well as the toxic solvent DMF were each used in separate syntheses of Compound **4** at different temperatures. The reaction solvent was also used to rinse the resin. Table 1 shows the results of these syntheses.

Table 1 Results from the Synthesis of Compound **4**

Solvent	Temperature (°C)	Yield (%)
DMF	25	85
DMF	75	80
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PC	25	95
PC	75	84

The successful synthesis of Compound **4** using nontoxic solvents encouraged researchers to pursue the synthesis of a larger, biologically active peptide. Bradykinin, a vasodilator containing nine amino acids (Arg-Pro-Pro-Gly-Phe-Ser-Pro-Phe-Arg), was synthesized using the same solid-state peptide synthesis protocol used to synthesize Compound **4**. Separate syntheses were performed using either DMF or PC, and the purity of bradykinin from

each synthesis was analyzed by high-performance liquid chromatography (HPLC) (Figure 2).

In each chromatogram, peak 1 has the same retention time as commercially obtained bradykinin, and the other peaks correspond to impurities. In chromatogram A, the impurity has a similar retention time to the product peak whereas the impurity peaks in chromatogram B have distinct retention times compared to the product peak.

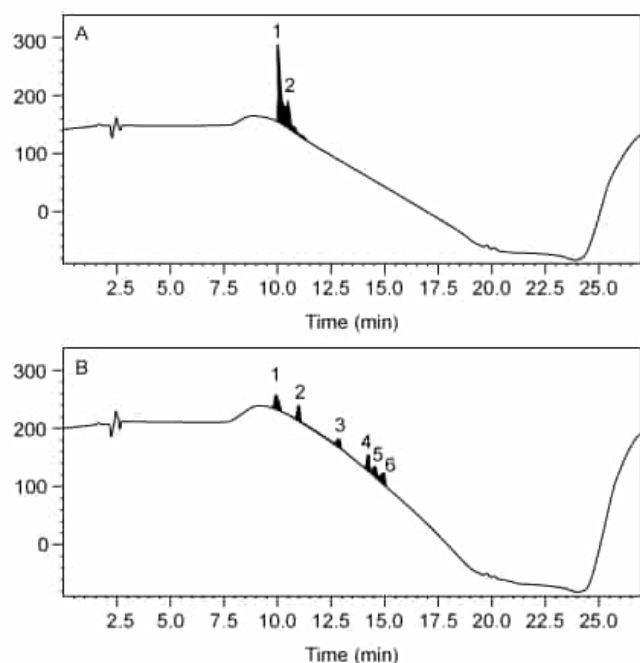


Figure 2 HPLC of synthetic bradykinin: (A) synthesis in DMF; (B) synthesis in PC

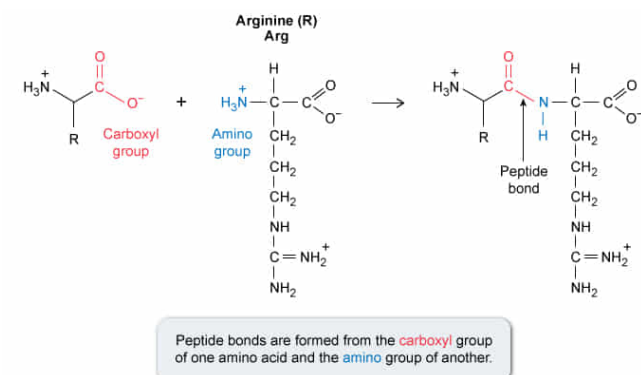
A peptide bond is formed in bradykinin synthesis when the amino group of arginine reacts with:

- ☐ A. an anhydride group. [1%]
- ✓ ☒ B. a carboxyl group. [86%]
- ☐ C. an amide group. [7%]
- ☐ D. a hydroxyl group. [5%]

Correct

86% answered correctly

Explanation:



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Amino acids are made up of an amino group, a carboxyl group, and a side chain, all bonded to the same carbon atom (known as the α -carbon).
 Peptides contain two or more amino acids bonded together. The amino group of one amino acid reacts with the carboxyl group of another amino acid to form a peptide bond.

The passage shows that the amino acid arginine (Arg, R) is located at both the **N- and C-termini** of bradykinin. The **amino group** of the arginine at the **C-terminus** reacts with the **carboxyl group** of the adjacent amino acid, phenylalanine (Phe, F), to form a **peptide bond**.

(Choice A) **Anhydride** groups do not react with amines to form peptide bonds, nor is this functional group present in any of the **20 common amino acids**.

(Choice C) An **amide** is *formed* during peptide bond formation rather than *reacting* with an amino group to form a peptide bond.

(Choice D) Although some amino acids contain hydroxyl groups in their side chains, this functional group does not react with an amine to form a peptide bond.

Educational objective:

Peptides contain two or more amino acids bonded together. The amino group of one amino acid reacts with the carboxyl group of another amino acid to form a peptide bond.

Time spent: 0 sec
 Subject:
 Organic Chemistry

QID: 402134
 System:
 Functional Groups and Their Reactions

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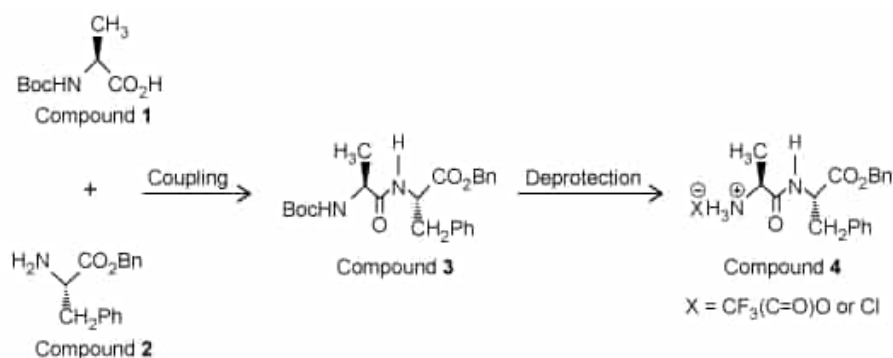


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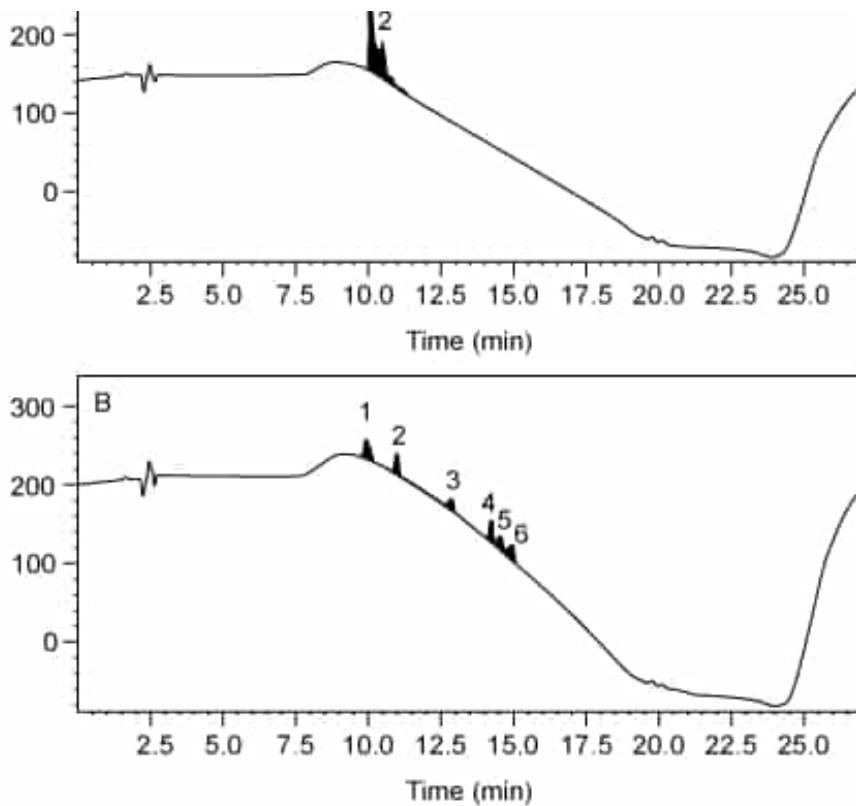


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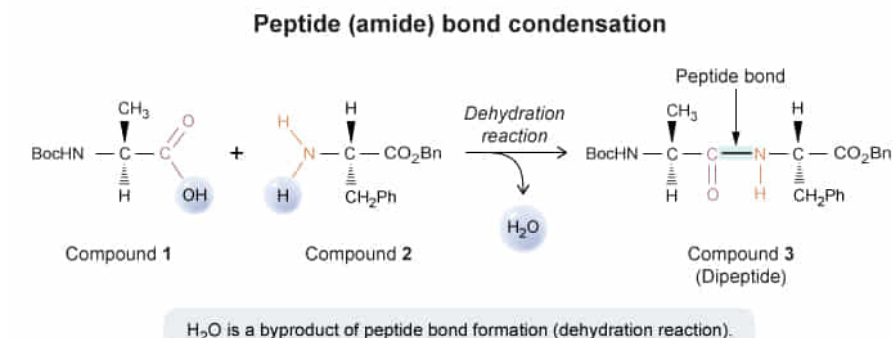
What is the byproduct of the formation of Compound **3** described in the passage?

- ✔ ☒ A. H_2O ; peptide bond formation is a dehydration reaction. [83%]
- ☐ B. H_2O ; peptide bond formation is a hydrolysis reaction. [8%]
- ☐ C. CO_2 ; peptide bond formation is a decarboxylation reaction. [4%]
- ☐ D. CO_2 ; peptide bond formation is a coupling reaction. [4%]

Correct

83% answered correctly

Explanation:



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Peptides are made up of two or more amino acids linked together through an amide bond known as a peptide bond. The amino group of one amino acid reacts with the carboxyl group of another amino acid to **form a peptide bond** through a **dehydration reaction**. The carboxyl group of one amino acid loses a hydroxyl group ($-\text{OH}$), and the amino group of the other loses a hydrogen atom, releasing **H_2O** as a **byproduct**.

Compound **3** is described as a dipeptide in the passage and is formed from Compounds **1** and **2** (both modified amino acids). The carboxyl group of Compound **1** and the amino group of Compound **2** undergo a dehydration reaction to form a peptide bond and release H_2O as a byproduct.

(Choice B) H_2O is a byproduct, but peptide bond formation is a *dehydration* reaction rather than a

hydrolysis reaction because water is *released* upon bond formation. **Hydrolysis** requires the *addition* of H_2O across a peptide bond, causing peptide bond cleavage.

(Choices C and D) The formation of a peptide bond is a coupling reaction, but H_2O is released rather than CO_2 . CO_2 is released in a decarboxylation reaction.

Educational objective:

Peptide bonds are formed through a dehydration reaction in which the carboxyl group of one amino acid loses a hydroxyl group and the amino group of another amino acid loses a hydrogen atom. H_2O is released as a byproduct in a dehydration reaction.

Time spent:0 sec

Subject:
Organic Chemistry

QID:402135

System:
Functional Groups and Their
Reactions

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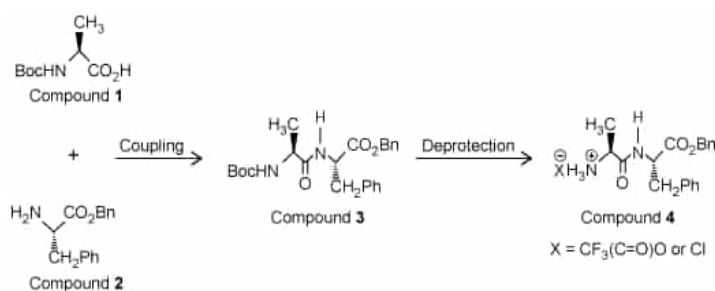


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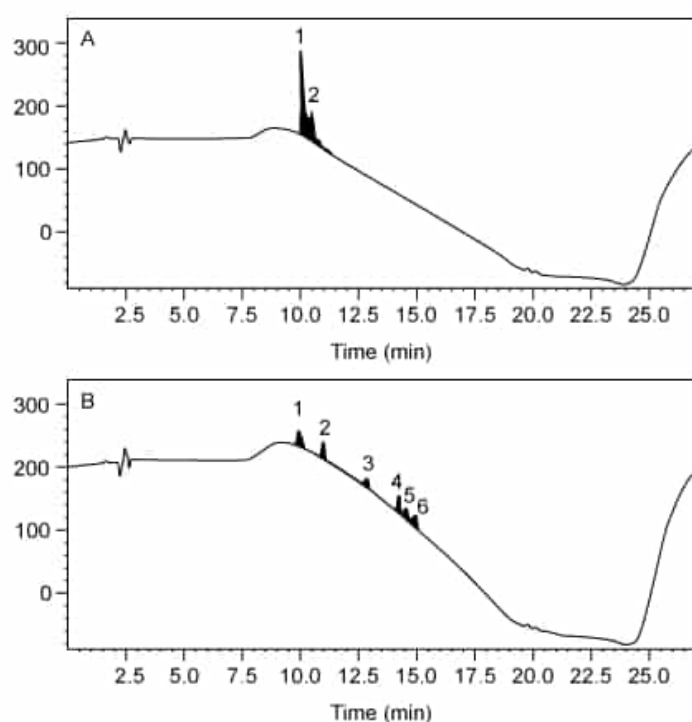


Figure 2 HPLC of synthetic bradykinin: (A) synthesis in DMF; (B) synthesis in PC

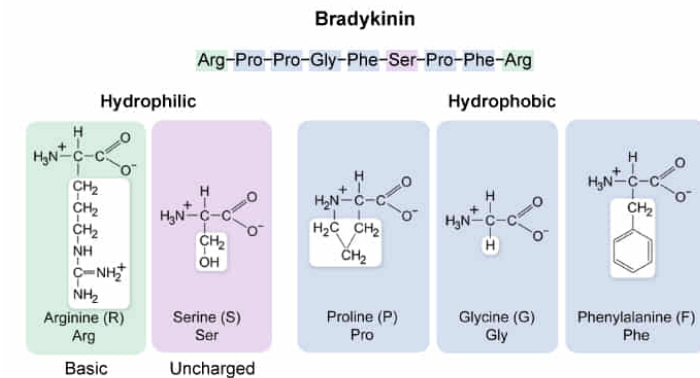
How many hydrophilic amino acid residues are present in bradykinin?

- ☐ A. 0 [2%]
- ✓ ☒ B. 3 [80%]
- ☐ C. 5 [7%]
- ☐ D. 6 [9%]

Correct

80% answered correctly

Explanation:



There are **three hydrophilic** amino acids in bradykinin: 2 Arg and 1 Ser.

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Amino acids all contain an amino group, a carboxyl group, and a side chain bonded to the same carbon (the α -carbon). The various amino acids differ only in the structure of their **side chains**, which contain different **functional groups** that give the amino acids their specific characteristics. This allows for the classification of amino acids based on side chain characteristics. Major **classifications** include charged (acidic or basic), hydrophobic (nonpolar), and hydrophilic (polar), and some amino acids can fall into **more than one** classification.

The passage states that the peptide bradykinin contains nine amino acids in the following sequence: Arg-Pro-Pro-Gly-Phe-Ser-Pro-Phe-Arg. The unique amino acids within this sequence are arginine (Arg, R), proline (Pro, P), glycine (Gly, G), phenylalanine (Phe, F), and serine (Ser, S). Arginine (Arg, R) has a positively charged guanidinium side chain, which is **basic**, and serine (Ser, S) has a **polar uncharged** side chain. Both residues are considered **hydrophilic**. The other residues (ie, Pro, Gly, Phe) contain an alkyl group, a hydride, and a phenyl group side chain, respectively. Each of these side chains is classified as **hydrophobic** (nonpolar).

Because there are two arginine residues and one serine residue, bradykinin has a total of **three hydrophilic residues**.

(Choice A) Serine and arginine are both hydrophilic, so bradykinin

does not have zero hydrophilic residues.

(Choice C) The two arginine residues and one serine residue add up to three hydrophilic residues, not five.

(Choice D) Bradykinin contains six *hydrophobic* residues (3 Pro, 2 Phe, 1 Gly), not six *hydrophilic* residues.

Educational objective:

Amino acids are made up of an amino group, a carboxyl group, and a side chain, all bonded to the α -carbon. Side chains are the point of difference in amino acid structure and contain different functional groups, which allows for amino acid classification based on the functional group characteristics. The main classifications include hydrophilic (including charged and polar uncharged) and hydrophobic.

Time spent:0 sec

Subject:
Organic Chemistry

QID:402136

System:
Functional Groups and Their Reactions

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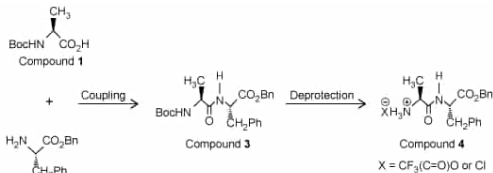


Figure 1 Synthesis of Compound **4**

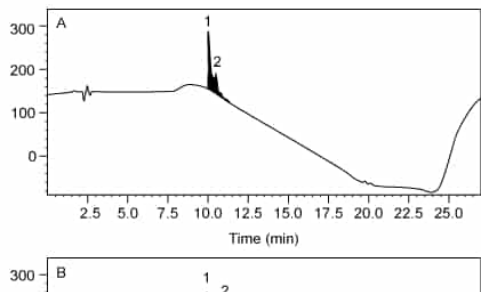
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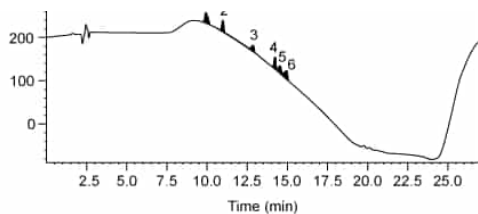
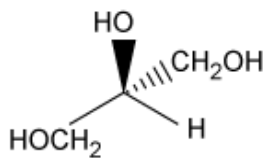
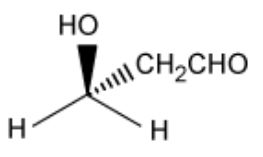
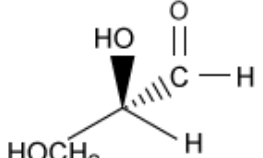
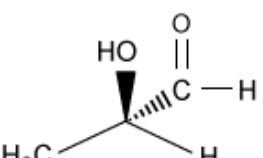


Figure 2 HPLC of synthetic bradykinin: (A) synthesis in DMF; (B) synthesis in PC

The L designation of the amino acids used in peptide synthesis is based on the absolute configuration of L-glyceraldehyde. What is the structure of L-glyceraldehyde?

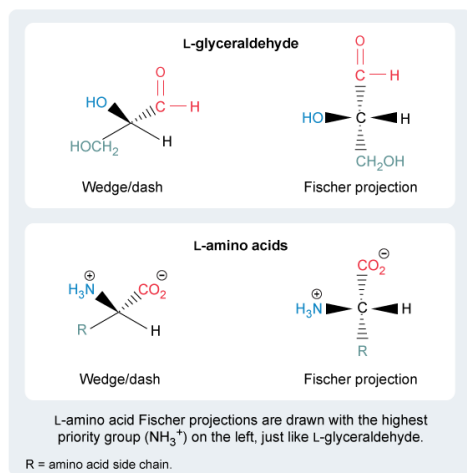
- ☐ A.
- 
- ☐ B.
- 
- ☒ C.
- 
- ☐ D.
- 

Correct

56% answered correctly

Explanation:

Configuration of L-amino acids based on L-glyceraldehyde



Amino acids have a stereocenter at the α-carbon because four different substituents are attached (an

amino group, a carboxyl group, a hydrogen atom, and an R group). The exception is glycine because it has only three unique substituents on the α -carbon rather than four (two hydrogen atoms, one of them being the side chain).

Two systems exist for labeling the absolute configuration of amino acids: the D/L system and the *R/S* system. The D/L system is based on the absolute configuration of D- and L-glyceraldehyde. Amino acids that resemble the configuration of L-glyceraldehyde are designated as L-amino acids, and those that resemble D-glyceraldehyde are D-amino acids. L-amino acids predominate in nature.

Glyceraldehyde is an oxidized form of glycerol. It is a three-carbon aldose with one aldehyde group and two hydroxyl groups. In Figure 1, Compound 1 is an L-amino acid with the R group at the top of the molecule shown as a wedge (ie, coming out of the plane). L-glyceraldehyde has a similar configuration; therefore, it should also have a functional group (the hydroxyl group) at the top of the molecule shown as a wedge.

(Choice A) This structure is glycerol and contains three hydroxyl groups (ie, one on each carbon). One of the terminal hydroxyl groups would need to be oxidized to an aldehyde to form glyceraldehyde. In addition, glycerol has no stereocenters and is neither D nor L.

(Choices B and D) Both of these three-carbon structures contain an aldehyde and one hydroxyl group. However, an aldehyde and two hydroxyl groups are present in glyceraldehyde.

Educational objective:

Glyceraldehyde is an aldose that can be derived from the oxidation of glycerol; it contains an aldehyde and two hydroxyl groups. D/L designations for all other molecules are based on D- and L-glyceraldehyde.

Time spent: 0 sec

Subject:
Organic Chemistry

QID: 402137

System:
Functional Groups and Their Reactions

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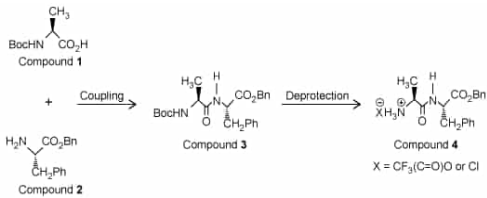


Figure 1 Synthesis of Compound **4**

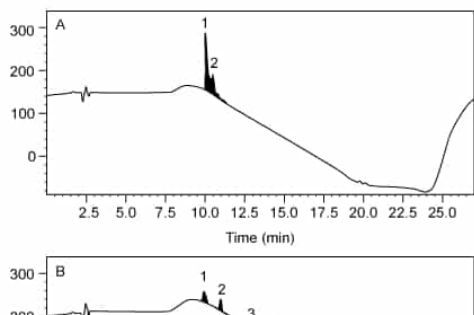
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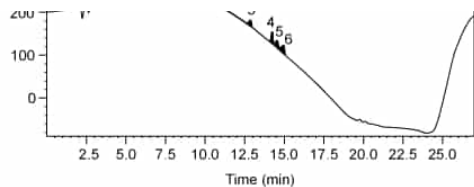


Figure 2 HPLC of synthetic bradykinin: (A) synthesis in DMF; (B) synthesis in PC

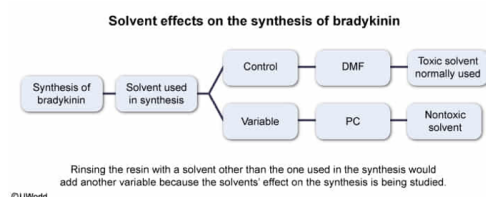
The most likely reason the resin was rinsed with the same solvent used for the reaction was to:

- ☒ A. avoid introducing another variable in the synthesis. [51%]
- ☐ B. decrease the amount of waste produced in the synthesis. [8%]
- ☐ C. prevent the occurrence of side reactions. [37%]
- ☐ D. speed up the peptide synthesis. [2%]

Correct

51% answered correctly

Explanation:



A **variable** is a component of an experiment that is **changed**. To determine how the variable affects the experimental results, all **other components** of the experiment must be **held constant**.

In the solid-phase peptide synthesis protocol described in the passage, the variable was the solvent used. Synthesis of bradykinin was carried out in two different solvents, DMF and PC, to determine which gave better results. If a different solvent was used in the rinse step than in the reaction step, the synthesis results could not be solely attributed to the solvent in which the reaction occurred.

(Choice B) The type of solvent used to rinse the resin does not affect how much waste is created during peptide synthesis. The *volume* of solvent used to rinse the resin affects the amount of waste created.

(Choice C) DMF, PC, and EC are all polar aprotic solvents, and are unlikely to produce different side reactions whether they were used in the initial reaction or not. For example, rinsing the DMF reaction with PC, or rinsing the PC reaction with EC, would not produce side reactions but would introduce another variable.

(Choice D) The rate of peptide synthesis does not depend on the solvent used in the rinse step. The resin must be rinsed regardless of the type of solvent used.

Educational objective:

Variables are components of an experiment that change while other components of the experiment are held constant. These parameters must be isolated to determine the effect a variable has on the results of an experiment.

Time spent: 0 sec

Subject:
Organic Chemistry

QID: 402138

System:
Functional Groups and Their Reactions

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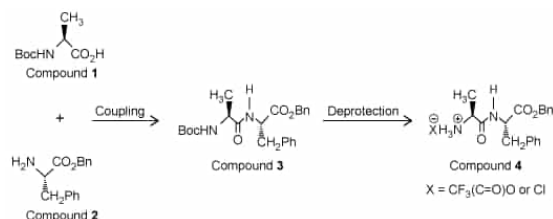


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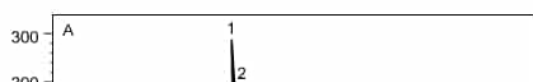
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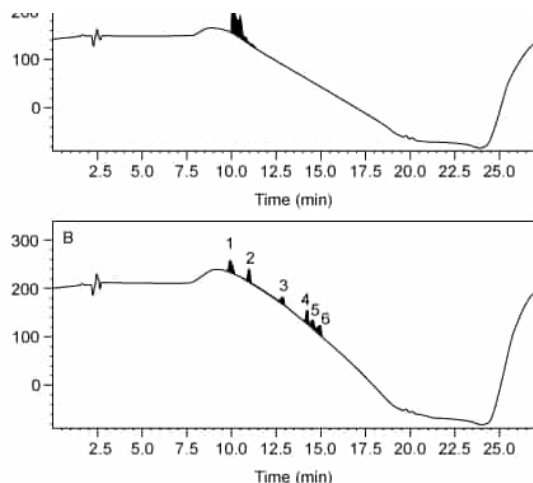


Figure 2 HPLC of synthetic bradykinin: (A) synthesis in DMF; (B) synthesis in PC

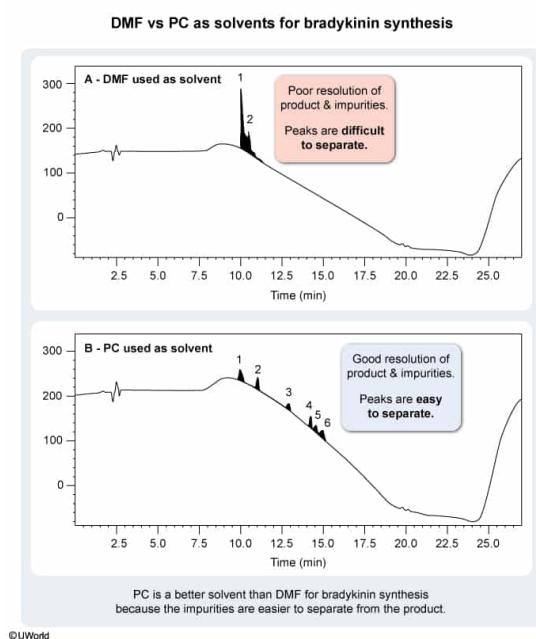
Why is PC a better solvent than DMF for peptide synthesis? Results of the peptide syntheses as described in the passage indicate that when PC is used as a solvent:

- ☐ A. the yield doubles compared to using DMF in the synthesis of the dipeptide. [2%]
- ☒ B. impurities are more easily separated from the final product in the synthesis of bradykinin. [93%]
- ☐ C. the yield decreases compared to using DMF in the synthesis of the dipeptide. [2%]
- ☐ D. no impurities are present in the synthesis of bradykinin. [1%]

Correct

93% answered correctly

Explanation:



To optimize reaction conditions for a synthesis reaction, the **methodology** should be evaluated. Various aspects of the **results** should be **considered**, including **yield** and **ease of purification**. Conditions that give the highest yield and easiest purification of the desired product are considered optimal.

HPLC chromatograms in Figure 2 show the results from the synthesis of bradykinin in DMF (A) and PC (B). Chromatogram A shows only one impurity peak (peak 2) whereas chromatogram B shows multiple impurities (peaks 2–6). Even though there was only one impurity when DMF was used, the impurity and product have very close retention times, causing **poor resolution** of the peaks and making separation difficult. There were more impurities when PC was used, but the resolution of peak 1 and the impurity peaks is good, making the **separation of bradykinin** and impurities **easier in PC** than in DMF. For this reason, **PC is a better solvent**.

(Choices A and C) Table 1 shows that the yield *increases* when PC is used as the solvent (compared to DMF), but the yield goes from 85% to 95% at 25 °C and from 80% to 84% at 75 °C. The yield is *not* doubled.

(Choice D) If there were no impurities from the synthesis, the only peak observed in the chromatogram would be the product peak (peak 1 in Figure 2). However, the chromatograms in Figure 2 show multiple peaks, indicating impurities are present in both syntheses.

Educational objective:

The methodology of a synthesis can be optimized by an analysis of the results, including yield and ease of purification. The conditions that give high product yield and easy separation of product from impurities are considered optimal.

Time spent:0 sec

Subject:
Organic Chemistry

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System:
Functional Groups and Their Reactions